TMPS Addendum V5 – Comprehensive Algorithmic & Implementation Annex

Supersedes V4 – incorporates error-handling, transfer-learning hooks and scalability notes.

0. Glossary & Helper Primitives

Symbol / Fn	Definition
ε	1×10^{-9} (divide-by-zero guard)
LOG_SUM_EXP(v)	m = max(v); m + ln Σ exp(v − m)
SIGMOID(x)	1 / (1 + e^(−x))
CLAMP(x,a,b)	max(a, min(x, b))
GAE(τ, V, γ, λ)	Generalised Advantage Estimation (Schulman 2016)
PROB_RATIO(π, π_0)	π(a
ONE_HOT(i,n)	length-n vector with 1 at index i

All helper primitives are implemented in the reference code library `tmps_core/helpers.py` .

1. Dynamic Ensemble Weighting

1.1 Diversity Metric

For models *i,j* over a sliding window **M = 64** predictions:

```
div[i][j] = 1 − |ρᵢⱼ| ,
ρᵢⱼ = PearsonCorr(predᵢ[-M:], predⱼ[-M:])
```

Error Handling:
If **M < 16** due to sensor dropouts, fallback to a **fixed equal-weight** strategy and raise an audit log

entry `"WARN_DIVERSITY_DEGRADED"` .

1.2 Weight Computation

```
def determine_weights(hist_err, conf, diversity, horizon, β_rl):
    reliability = 1.0 / (np.array(hist_err) + ε)

    logits = np.log(reliability + ε)
    reliability = np.exp(logits - log_sum_exp(logits))

    conf = conf / (conf.sum() + ε)

    β = get_beta(β_rl, horizon)
    w = β*reliability + (1-β)*conf

    w *= 1 + 0.15 * diversity.mean(axis=1)     # diversity bonus
    w /= w.sum()

    return w
```

Transfer-Learning Hook:
When a **new hardware node** joins, initialise `hist_err` with priors transferred from the *fleet knowledge base* via:

```
hist_err = load_priors(node_arch_id)  # ≤ Section VII-1
```

2. Criticality-Aware Precision Scaling

2.1 Criticality Score

```
criticality = max(
    task_meta,                    # LOW=0.2 | MED=0.5 | HIGH=0.9
    latency/latency_SLA,
    user_override or 0
)
criticality = CLAMP(criticality, 0, 1)
```

2.2 Composite Score & Precision

```
def composite_score(pred_T, crit, thr_cnt, energy):
    t_risk   = sigmoid((pred_T - 70)/5)
    th_pen   = min(thr_cnt/10, 1)
    e_risk   = 1 - energy

    return 0.45*crit + 0.30*(0.6*t_risk+0.4*th_pen) + 0.15*e_risk
```

Precision tier selection + hysteresis identical to V4; error path returns **FP32** if any temperature sensor is stale > 2 s.

2.3 Hardware Mapping

Tier	CPU	GPU	NPU / ASIC
HIGH	FP32 FMA	CUDA FP32	float32
MID	AVX-FP16/BF16	TensorCore FP16/BF16	bfloat16
LOW	AVX-VNNI INT8	TensorCore INT8	int8

3. PPO-Based Self-Optimisation

3.1 RL Formalism

State **S (12-D)** adds `sensor_health` & `fleet_prior_delta` to V4's 10-D vector.
Action **A** unchanged.
Reward **R** now:

```
R = R₀ + 0.2·R_aux − 0.3·sensor_drop_penalty
```

`sensor_drop_penalty = 1` if any mandatory sensor stale, else 0.

3.2 Transfer-Learning

When a node boots with no local PPO checkpoint:

```
θ_actor, θ_critic ← fetch_fleet_checkpoint(similar_arch)
```

Weights are fine-tuned online; meta-data logged for continual learning.

4. Cross-Module Coordination

Updated sequence diagram includes **Error Bus** for sensor faults and **Fleet KB** for prior sharing (see Figure 2-B).

5. Structured Decision Log

Adds `"sensor_ok": bool` , `"transfer_source": str|null` .

6. Benchmark Slice (v2)

Metric	Unit	Workload	Baseline (DVFS)	TMPS V5	Δ
MAE (5 min)	°C	ResNet-50	4.9	1.4	–71 %
Tj_peak	°C	ResNet-50	93	86	–7.5 %
Throttle Time	%	ResNet-50	12.4	2.5	–79 %
Energy / Task	kJ	ResNet-50	1.74	1.42	–18 %
Hi-Crit TPS	ops/s	Mix	8 950	9 680	+8.1 %

7. Scalability & Complexity Notes

1. Ensemble update $O(N^2)$ for diversity; $N \leq 5 \rightarrow$ negligible.
2. PPO inference: < 0.3 ms on Ryzen 7950X.
3. Distributed fleet: gRPC async calls; bandwidth ≈ 2 KB/s/node.
4. Error-Handling Strategy

Failure	Mitigation
Sensor stale > 2 s	Raise `sensor_drop_penalty` ; force FP32; log event
Diversity window < 16	Equal-weight ensemble; disable diversity bonus
PPO reward < –0.1 for 10 iters	Reload heuristic policy

End of Addendum V5